

MAP

HASHMAP:

```
import java.util.*;
public class Main
{
    public static void main(String[] args)
    {
        HashMap<Integer, String> map = new HashMap<>();
        map.put(101, "Rose");
        map.put(102, "Cha Eun-woo");
        map.put(103, "Sana");
        map.put(104, "Felix");
        System.out.println("After put(): " + map);
        System.out.println("Value of key 102: " + map.get(102));
        System.out.println("Contains key 103? " + map.containsKey(103));
        System.out.println("Contains value 'Sana'? " + map.containsValue("Sana"));
        System.out.println("Size: " + map.size());
        System.out.println("Keys: " + map.keySet());
        System.out.println("Values: " + map.values());
        System.out.println("Entries: " + map.entrySet());
        map.remove(104);
        System.out.println("After remove(): " + map);
        System.out.println("Is Empty? " + map.isEmpty());
        map.clear();
        System.out.println("After clear(): " + map);
    }
}
```

OUTPUT:

```
After put(): {101=Rose, 102=Cha Eun-woo, 103=Sana, 104=Felix}
Value of key 102: Cha Eun-woo
Contains key 103? true
Contains value 'Sana'? true
Size: 4
Keys: [101, 102, 103, 104]
Values: [Rose, Cha Eun-woo, Sana, Felix]
Entries: [101=Rose, 102=Cha Eun-woo, 103=Sana, 104=Felix]
After remove(): {101=Rose, 102=Cha Eun-woo, 103=Sana}
Is Empty? false
After clear(): {}
```

COUNT FREQUENCY OF CHARACTERS

```
import java.util.*;
public class Main
```

```

{
    public static void main(String[] args)
    {
        String str = "banana";
        HashMap<Character, Integer> map = new HashMap<>();
        for (char ch : str.toCharArray())
        {
            map.put(ch, map.getOrDefault(ch, 0) + 1);
        }
        System.out.println(map);
    }
}

```

OUTPUT:

{a=3, b=1, n=2}

FIRST NON REPEATRED ELEMENT:

```

import java.util.*;
public class Main
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the size of the array: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        HashMap<Integer, Integer> map = new HashMap<>();
        System.out.println("Enter the array elements:");
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
            map.put(arr[i], map.getOrDefault(arr[i], 0) + 1);
        }
        boolean found = false;
        for (int i = 0; i < n; i++)
        {
            if (map.get(arr[i]) == 1)
            {
                System.out.println("First Non-Repeated Element: " + arr[i]);
                found = true;
                break;
            }
        }
        if (!found)
        {

```

```

        System.out.println("No non-repeated element found.");
    }
}
}

```

OUTPUT:

Enter the size of the array: 8
Enter the array elements:
1 2 2 1 4 5 5 6
First Non-Repeated Element: 4

ANAGRAM:

```

import java.util.*;
public class Main
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first string: ");
        String str1 = sc.nextLine().toLowerCase();
        System.out.print("Enter second string: ");
        String str2 = sc.nextLine().toLowerCase();
        if (str1.length() != str2.length())
        {
            System.out.println("Not Anagram");
            return;
        }
        HashMap<Character, Integer> map = new HashMap<>();
        for (char ch : str1.toCharArray())
        {
            map.put(ch, map.getOrDefault(ch, 0) + 1);
        }
        for (char ch : str2.toCharArray())
        {
            if (!map.containsKey(ch))
            {
                System.out.println("Not Anagram");
                return;
            }
            map.put(ch, map.get(ch) - 1);
            if (map.get(ch) == 0)
            {
                map.remove(ch);
            }
        }
    }
}

```

```
    }  
    if (map.isEmpty())  
    {  
        System.out.println("Anagram");  
    }  
    else  
    {  
        System.out.println("Not Anagram");  
    }  
}  
}
```

OUTPUT:

Enter first string: listen

Enter second string: silent

Anagram